

LA HW

Included exercises: 1, 9, 19, 21, 26, 29, 39, 45, 50-51, 54-55, 60, 62-63

Exercises 1-8

In Exercises 1–8, determine whether each subset of $\mathcal{M}_{2 \times 2}$ is linearly independent or linearly dependent.

$$\left\{ \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}, \begin{bmatrix} 1 & -5 \\ -4 & 0 \end{bmatrix}, \begin{bmatrix} 3 & -1 \\ 2 & 2 \end{bmatrix} \right\}$$

Original Exercise 1

linearly independent or linearly dependent.

$$\left\{ \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}, \begin{bmatrix} 1 & -5 \\ -4 & 0 \end{bmatrix}, \begin{bmatrix} 3 & -1 \\ 2 & 2 \end{bmatrix} \right\}$$

Exercises 9-16

In Exercises 9–16, determine whether each subset of $\mathcal{P}$ is linearly independent or linearly dependent.

Original Exercise 9

linearly independent or linearly dependent.

$$\{1 + x, 1 - x, 1 + x + x^2, 1 + x - x^2\}$$

Exercises 17-24

In Exercises 17–24, determine whether each subset of $\mathcal{F}(\mathbb{R})$ is linearly independent or linearly dependent.

Original Exercise 19

$$\{\sin t, \sin^2 t, \cos^2 t, 1\}$$

Original Exercise 21

$$\{e^t, e^{2t}, \dots, e^{nt}, \dots\}$$

Exercises 25-30

linearly independent or linearly dependent
 $(5, -2), (5, 1), (5, -5), (5, 5), (5, -1)$

Original Exercise 26

$n = 3$; $(1, 8)$, $(2, 5)$, and $(3, -4)$

Original Exercise 29

$n = 4$; $(-1, 5)$, $(0, 2)$, $(1, -1)$, $(2, 2)$

Exercises 31-48

In Exercises 31–48, determine whether the statements are true or false.

Original Exercise 39

In $\mathcal{P}$, the set $\{x, x^3, x^5, \dots\}$ is linearly dependent.

Original Exercise 45

of V and W are equal.

The dimension of $\mathcal{L}(\mathcal{R}^n, \mathcal{R}^m)$ is $m + n$.

Original Exercise 50

early independent. Justify your answer.

Exercises 51-58

In Exercises 51–58, find a basis for the subspace W of the vector space V .

Original Exercise 51

V .

Let W be the subspace of symmetric 3×3 matrices and

Original Exercise 54

Let W be the subspace of $V = \mathcal{P}_n$ consisting of polyno-

Original Exercise 55

Let W be the subspace of $V = \mathcal{P}_n$ consisting of polyno-

Exercises 60-65

The following definitions and notation apply to Exercises 60–65:

Definition An $n \times n$ matrix is called a **magic square of order**

Original Exercise 60

equal to 0.

(a) Show that V_n is a subspace of $\mathcal{M}_{n \times n}$.

Original Exercise 62

square B in W_n such that $A = B + sC_n$.

Original Exercise 63

Prove that V_3 has dimension equal to 3.